Lecture 15
Random Variables and Discrete Probability Distributions

Review

1 What is a sample space?

The collection of all possible outcomes of a random trial.

2 What is the law of large numbers?

As the number of trials grows, the relative frequency of A gets closer and closer to P(A).

3 What are disjoint (mutually exclusive) events?

Events with no outcomes in common, so they cannot happen together: $A \cap B = \emptyset$

Two more rules

general addition rule

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$: subtract the overlap so it is not counted twice

multiplication rule

if A and B are independent (one occurring does not change the probability of the other): $P(A \cap B) = P(A) \cdot P(B)$

disjoint ≠ independent

disjoint events cannot happen together, so knowing one happened tells you the other did not

Example: one die

Roll a fair six-sided die once.

S = {1, 2, 3, 4, 5, 6}, each outcome has probability 1/6

(a) What is P(not a six)?
(b) What is P(1 ∪ 6)?
(c) What is P(even ∪ greater than 3)?

(a) 1 – 1/6 = 5/6 (rule 4)
(b) 1/6 + 1/6 = 1/3 (rule 3: a 1 and a 6 are disjoint)
(c) 3/6 + 3/6 – 2/6 = 4/6 (the 4 and the 6 are in both)

Example: two dice

Roll a fair die twice. The two rolls are independent.

P(six) = 1/6 on each roll

(a) What is P(six on both rolls)?
(b) What is P(at least one six)?

(a) (1/6)(1/6) = 1/36 (multiplication rule)
(b) 1 – P(no sixes) = 1 – (5/6)(5/6) = 11/36 ≈ 0.31

Data: student heights

A survey of 379 college students recorded each student's height and gender.

X = the height of a student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
…	…

n = 379 students (the file lists the women first, then the men, each sorted by height)

Example: unions and intersections

Pick one of the 379 students at random. F = female, T = at least 70 in.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

(a) What is P(F ∩ T)?

(b) What is P(F ∪ T)?

(a) 19 / 379 ≈ 0.05

(b) 262/379 + 106/379 – 19/379 = 349/379 ≈ 0.92

Are F and T independent?

Pick one of the 379 students at random. F = female, T = at least 70 in.

	At least 70 in	Under 70 in	Total
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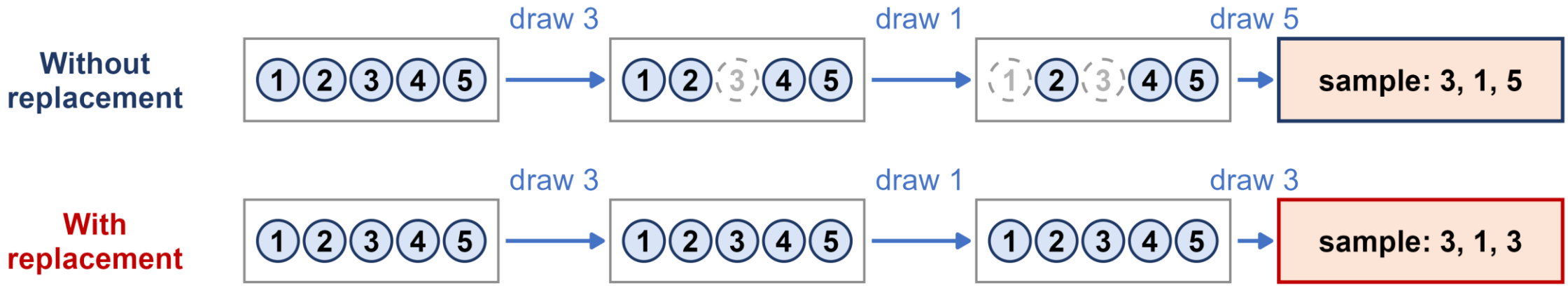
Does knowing a student is female change the chance they are at least 70 in?

Yes: 106/379 ≈ 28% of all students, but only 19/262 ≈ 7% of the women

Not independent: $P(F) \cdot P(T) \approx 0.69 \cdot 0.28 \approx 0.19$, but $P(F \cap T) \approx 0.05$

With or without replacement?

Week 4: drawing a sample of 3 balls from a bag of 5, two ways.



Are the draws independent?

With replacement: yes. The bag is the same on every draw, so $P(A \cap B) = P(A) \cdot P(B)$
Without replacement: no. Each draw changes what is left, so the next probability depends on the last draw

Example: drawing from an urn

An urn holds 3 red, 4 blue and 3 yellow balls. Draw 2 balls without replacement.

10 balls in all: 3 red, 4 blue, 3 yellow

(a) What is P(both red)?
(b) What is P(no red balls)?
(c) What is P(both the same color)?

(a) (3/10)(2/9) = 6/90 ≈ 0.067
(b) (7/10)(6/9) = 42/90 ≈ 0.467
(c) (3/10)(2/9) + (4/10)(3/9) + (3/10)(2/9) = 24/90 ≈ 0.267

Example: drawing two aces

A standard deck has 52 cards, 4 of them aces. Draw two cards.

P(ace on the first draw) = 4/52

(a) What is P(two aces) if the first card is put back (with replacement)?
(b) What is P(two aces) if it is not put back (without replacement)?

(a) (4/52)(4/52) = 1/169 ≈ 0.0059: the draws are independent
(b) (4/52)(3/51) = 1/221 ≈ 0.0045: only 3 aces are left among 51 cards

Random variables

random variable

the outcome of a random trial recorded as a number, written with a capital letter such as X

discrete

its possible values can be listed: the number of heads in 3 tosses, X = 0, 1, 2, 3

continuous

it can take any value in an interval: a student's height

Example: three coin tosses

Toss a fair coin 3 times: 2 × 2 × 2 = 8 equally likely outcomes.

X = the number of heads

(a) List the sample space S.

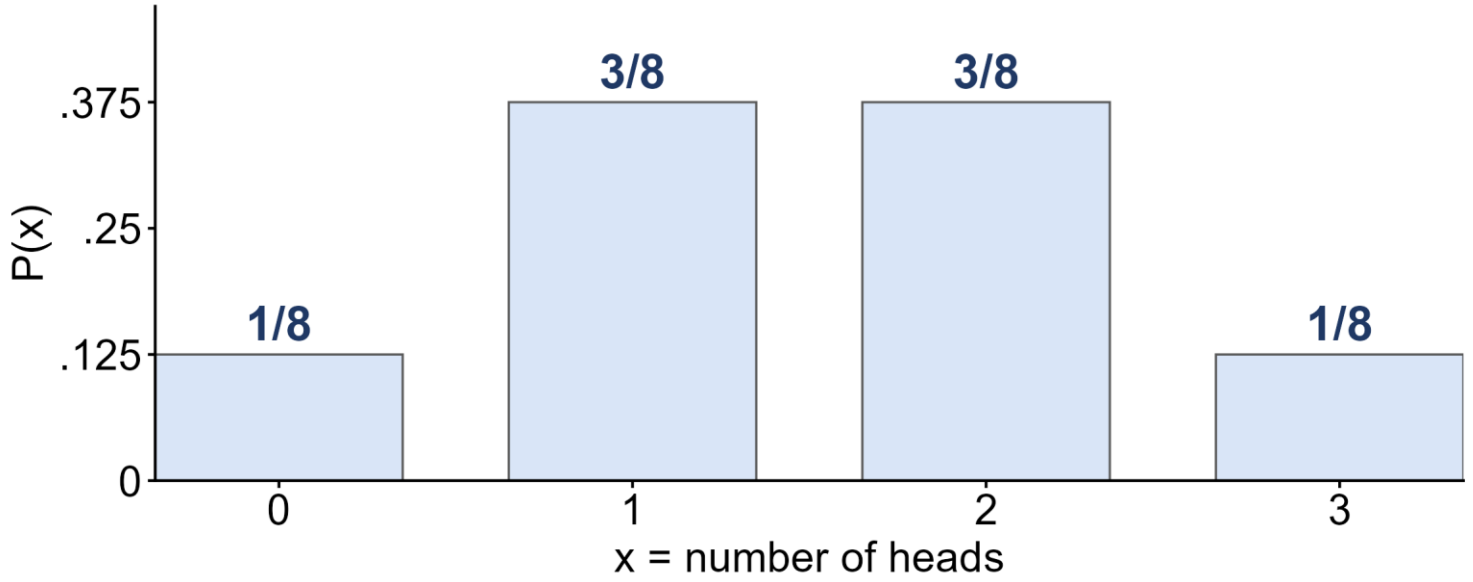
(b) How many outcomes give each value of X?

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

X = 0: 1 X = 1: 3 X = 2: 3 X = 3: 1 (out of 8)

Discrete probability distributions

probability distribution a function that assigns a probability to each possible value of X; each is between 0 and 1, and they add to 1



1 + 3 + 3 + 1 = 8, so the probabilities add to 8/8 = 1

Example: using a distribution

X = the number of heads in 3 tosses of a fair coin.

P(0) = 1/8 P(1) = 3/8 P(2) = 3/8 P(3) = 1/8

(a) What is P(X ≥ 2)?
(b) What is P(X ≤ 1)?
(c) What is P(1 ≤ X ≤ 2)?

(a) 3/8 + 1/8 = 4/8
(b) 1 – P(X ≥ 2) = 1 – 4/8 = 4/8
(c) 3/8 + 3/8 = 6/8